10Pearls Shine Intern

PROJECT REPORT

AQI Predictor — Serverless ML Pipeline

Submitted by

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1. Executive Summary

This report presents the design, development, and current status of the AQI Predictor — a fully serverless, end-to-end machine learning pipeline built to forecast the Air Quality Index (AQI) for the next three days. The project leverages real-world air quality data fetched from the AQICN API and employs a Random Forest Regressor model for accurate multi-day predictions.

The pipeline covers every stage of a production-grade ML system: automated data ingestion, feature engineering, model training with explainability via SHAP values, feature storage on Hopsworks, and a Streamlit-based interactive dashboard accessible from anywhere. As of the submission date, the project is approximately 75% complete, with the core ML pipeline and dashboard fully operational.

2. Project Overview

2.1 Objective

The primary objective of this project is to predict the Air Quality Index (AQI) for an urban area over the next three consecutive days. The system operates in a fully automated, serverless manner — eliminating the need for any manual data collection or model retraining steps.

2.2 Approach

The project follows a systematic, end-to-end machine learning workflow:

- Data Ingestion — Live AQI data is retrieved automatically from the AQICN API.
- Feature Engineering — Lag features, rolling statistics, and time-based cyclical signals are extracted.
- Feature Store — Engineered features are stored and served via the Hopsworks feature store.
- Model Training — A Random Forest Regressor is trained on historical AQI data and registered.
- Prediction & Visualization — A 3-day AQI forecast is generated and displayed on a Streamlit dashboard.

2.3 Deliverables

- Fully automated ML data pipeline
- Trained and registered Random Forest model with SHAP explainability
- Interactive Streamlit dashboard with live predictions
- REST API backend via Flask (in progress)

3. Technology Stack

The project utilizes a modern, production-grade technology stack covering data engineering, machine learning, explainability, and deployment:

Technology	Category	Purpose
Python	Core Language	Primary programming language for all pipeline components
Scikit-learn	ML Library	Random Forest Regressor model training and evaluation
TensorFlow	Deep Learning	Deep learning framework (planned for future enhancements)
Hopsworks	Feature Store	Centralized feature storage, versioning, and serving
AQICN API	Data Source	Live real-time Air Quality Index data for target city
Apache Airflow / GitHub Actions	Orchestration	Automated pipeline scheduling and execution
Streamlit	Dashboard	Interactive web dashboard for AQI prediction display
Flask	Backend	REST API layer for serving model predictions (planned)
SHAP	Explainability	Model feature importance and explainability layer
Git	Version Control	Source code versioning and collaboration

4. ML Pipeline Architecture

The pipeline is designed as a fully serverless, modular architecture. Each stage can be independently updated or retrained without interrupting the end-to-end flow.

Stage 1 — Data Collection

Live AQI data is fetched automatically from the AQICN API for the configured target city. The API token is securely configured, and data collection runs on a scheduled basis via GitHub Actions.

Stage 2 — Feature Engineering

Raw AQI readings are transformed into a rich feature set for model training:

- AQI Lag Features (Lag 1 through Lag 7) — Previous days' AQI values capturing temporal patterns.
- Rolling Mean Features — 7-day and 14-day moving averages smoothing short-term noise.
- Time-Based Features — Hour of day and day of week encoded as cyclical signals.

Stage 3 — Feature Store (Hopsworks)

Engineered features are stored in the Hopsworks feature store, which enables version-controlled, consistent feature serving for both model training and real-time inference. The Hopsworks API token is configured and ready.

Stage 4 — Model Training

A Random Forest Regressor is trained on the historical feature dataset. The trained model is registered in the Hopsworks model registry, enabling versioned model management and easy rollback. SHAP (SHapley Additive exPlanations) values are computed to provide transparent feature importance explanations.

Stage 5 — Predictions & Dashboard

On each pipeline run, a 3-day AQI forecast is generated and pushed to the Streamlit dashboard. Users can view today's AQI, tomorrow's forecast, and a Day 3 prediction alongside a 7-day historical trend chart and SHAP feature importance visualization.

5. Model Details

5.1 Algorithm: Random Forest Regressor

The Random Forest Regressor was selected as the primary model based on the following considerations:

- Handles non-linear AQI patterns effectively without requiring complex tuning.
- Robust to missing or noisy sensor data — a common issue with air quality datasets.
- No feature scaling required, simplifying the preprocessing pipeline.
- Provides native feature importance scores, complementing SHAP values.
- Strong performance on time-series forecasting tasks with minimal overfitting risk.
- Ensemble approach reduces variance and improves generalization.

5.2 Key Features Used

- AQI Lag 1–7 — Previous 7 days of AQI readings (most impactful features per SHAP analysis).
- Rolling Mean 7d & 14d — Short-term and medium-term trend indicators.
- Hour of Day — Captures intra-day pollution cycles.
- Day of Week — Accounts for weekly traffic and industrial patterns.

5.3 Model Explainability

SHAP (SHapley Additive exPlanations) values are computed for each prediction to provide a transparent, interpretable explanation of the model output. The dashboard displays the top features by SHAP importance, allowing users to understand which factors are driving today's AQI forecast.

6. Streamlit Dashboard

The Streamlit dashboard serves as the primary user interface for the AQI Predictor. It is fully built and functional, displaying the following components:

- Today's AQI — Current day prediction with qualitative health category label (e.g., Moderate, Good).
- Tomorrow's Forecast — Next-day AQI prediction.
- Day 3 Forecast — Three-day-ahead prediction.
- AQI Trend Chart — A 7-day historical AQI chart combined with the 3-day forecast.
- SHAP Feature Importance — Bar chart showing the top contributing features to the current prediction.

Sample dashboard output (from pipeline run):

```
Today: AQI 82 (Moderate)   |   Tomorrow: AQI 67 (Good)   |   Day 3: AQI 91 (Moderate)
```

7. Project Progress & Status

The project is currently approximately 75% complete. The table below summarizes completed and remaining components:

✅ Completed	⏳ Remaining
AQICN API integration (token configured) Automated data pipeline / collection Feature engineering module Random Forest model training Hopsworks feature store (token configured) Streamlit dashboard — fully built Model registered in Hopsworks	Flask backend / REST API layer Flask + Streamlit integration Backend deployment (serverless) End-to-end integration testing

Overall Completion: 75%

8. Next Steps

To complete the remaining 25% of the project, the following tasks are planned:

- Develop the Flask REST API layer to expose model predictions as HTTP endpoints.
- Integrate the Flask backend with the Streamlit dashboard for unified data flow.
- Deploy the Flask backend as a serverless function (e.g., AWS Lambda or Google Cloud Run).
- Conduct end-to-end integration testing across the full pipeline to verify correctness.
- Optionally explore TensorFlow-based deep learning models as an enhancement.

9. Conclusion

The 10Pearls AQI Predictor project demonstrates the design and implementation of a production-ready, serverless machine learning pipeline for real-world air quality forecasting. The core ML pipeline — from data ingestion through model training to the Streamlit dashboard — is fully functional and ready for demonstration.

The use of Hopsworks as a feature store, SHAP for model explainability, and a fully automated orchestration pipeline positions this project as a comprehensive example of modern MLOps practices. Upon completion of the Flask backend and integration testing, the system will represent a fully deployable, end-to-end AQI forecasting solution.

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